

Stochastic Path Selection

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Reply-To: pzg8794@rit.edu
To: Daniel Krutz <dxkvse@rit.edu>

Wed, Nov 12, 2025 at 12:24 PM

Hey Krutz,

Thanks for sending this over!

I actually think this is a very solid idea. From a high level, it not only makes sense, it aligns almost perfectly with the direction I've already been pushing our current framework toward. In fact, the recent upgrades to our infrastructure were made specifically to support exactly this kind of evolution.

So rather than pivoting, I see this as the natural progression of what we've built. The current framework was intentionally structured as a testbed to support multiple interchangeable policy learners—making it ideal for testing EXA-GP-based decision agents alongside our neural and adversarially trained variants.

I'm fully on board with formalizing this direction. I'd also love to align on specific performance metrics and tradeoff objectives we'd want these evolved strategies to optimize for (e.g., fidelity vs. purification vs. coherence decay), and how they'd differ from real-time learners like iCMAB.

I really love how tightly this complements the work already in place.

Best,
Piter

On Wed, Nov 12, 2025 at 6:11AM Daniel Krutz <dxkvse@rit.edu> wrote:
Hi Piter.

I was hoping that you could answer a question for me. What do you think about this idea from a high level? Good, bad, obscenely stupid?

Basically, my goal is to determine if either new approach can adequately address a new problem in quantum path selection, or the ML aspect was designed to solve a unique quantum problem.

Quantum entanglement path selection is the process of dynamically identifying and optimizing routes through intermediate quantum nodes to establish high-fidelity entanglement between distant qubits in a quantum network. It is crucial that quantum networks make high-quality, adaptive routing decisions in order to establish entanglement across dynamic, noisy and resource-limited topological [Cite1]. Unfortunately, current state-of-the-art techniques are limited by their inability to be explainable, scalable and policy transferable [Cite2]. This is especially the case trying to balance competing objectives such as maximizing fidelity, minimizing purification overhead, and maintaining coherence lifetimes. The objective of this work is to address the problem of how to evolve interpretable, adaptive routing strategies that can efficiently learn and generalize across varying quantum network conditions.

We will address the issue of current techniques being unable to be explainable, scalable and policy transferable through the adaptation and modification of a hybrid evolutionary machine learning method that combines graph-based genetic programming (GP) with neuroevolution , specifically the EXA-GP framework. This model will provide the benefits of: Adaptive optimization, Explainability, Scalability, Tradeoff management, Robustness under uncertainty.

Cite 1
<https://tqe.ieee.org/2025/02/11/entanglement-routing-in-quantum-networks-a-comprehensive-survey/>
<https://arxiv.org/abs/2408.01234>

Cite 2
<https://link.springer.com/article/10.1007/s11128-024-04287-z>
<https://link.springer.com/article/10.1007/s11128-024-04287-z>